

Symmetries and Anomalies of Kitaev Spin- S Models: Identifying Symmetry-Enforced Exotic Quantum Matter

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Motivations

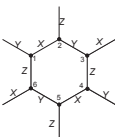
Model:

$$H = - \sum_{\langle ij \rangle} J_{\alpha ij} S_i^{\alpha ij} S_j^{\alpha ij}$$

Useful property

$$e^{i\pi S^\mu} e^{i\pi S^\nu} = \epsilon_{\mu\nu\lambda} e^{i\pi S^\lambda}, \quad e^{i\pi S^\mu} e^{i\pi S^\nu} = (-1)^{2S} e^{i\pi S^\nu} e^{i\pi S^\mu}$$

Conserved $\mathbb{Z}_2$ flux:

$$W_p = \prod_{j \in \partial p} e^{i\pi S_j^{\hat{n}_j}}$$


Observations: Even-odd effect (H. Ma, PRL 130, 156701 (2023))

- $S \in \mathbb{Z} + \frac{1}{2}$: gauge charge is **fermionic** and the $\mathbb{Z}_2$ gauge field is **deconfined**.
- $S \in \mathbb{Z}$: gauge charge is **bosonic** and the $\mathbb{Z}_2$ gauge field can be Higgsed.

Questions:

- To what extent is the even-odd effect stable against perturbation?
- Can the results be obtained via a more elementary, direct approach (via spins instead of partons)?

- (Global) $\mathbb{Z}_2^{[1]}$ 1-form symmetry: Generators are two global loops (on a torus)

$$\prod_{j \in \mathcal{L}_v} e^{i\pi S_j^{\hat{n}_j}} \quad , \quad \prod_{j \in \mathcal{L}_h} e^{i\pi S_j^{\hat{n}_j}}$$

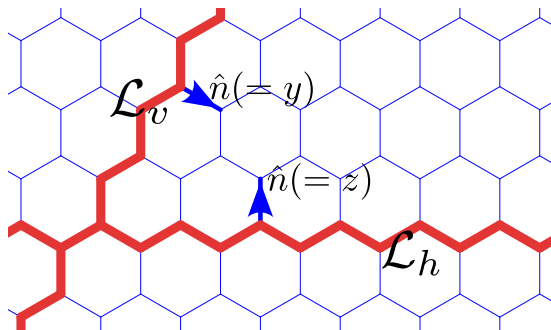
- $\mathbb{Z}_2^T$ time reversal symmetry: $S_i^\mu \rightarrow -S_i^\mu$
- $\mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$: Generators are

$$\prod_j e^{i\pi S_j^x} \quad , \quad \prod_j e^{i\pi S_j^y}$$

Notice that $e^{i\pi S_j^x} e^{i\pi S_j^y} = e^{i\pi S_j^z}$

Full internal symmetry: $\mathbb{Z}_2^{[1]} \times \mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$

Symmetries



$$\prod_{j \in \mathcal{L}_v} e^{i\pi S_j^{\hat{n}_j}} = \prod_{j \in \mathcal{L}_v} e^{i\pi S_j^y}, \quad \prod_{j \in \mathcal{L}_h} e^{i\pi S_j^{\hat{n}_j}} = \prod_{j \in \mathcal{L}_v} e^{i\pi S_j^z}$$

Anomalous 1-Form Symmetry

Anomaly: Presence of obstruction to gauging symmetries. Here we discuss the 1-form symmetry $\mathbb{Z}_2^{[1]}$.

- The action of a generator of $\mathbb{Z}_2^{[1]}$ symmetry

$$S_k^\mu \rightarrow \left(\prod_{j \in \mathcal{L}} e^{i\pi S_j^{\hat{n}j}} \right) S_k^\mu \left(\prod_{j \in \mathcal{L}} e^{i\pi S_j^{\hat{n}j}} \right)^\dagger = \lambda_k^\mu S_k^\mu$$

where

$$\lambda_k^\mu = \begin{cases} -1, & k \in \mathcal{L}, \mu \in \mathcal{L} \\ 1, & \text{otherwise.} \end{cases}$$

- Introduce the $\mathbb{Z}_2$ gauge field A_{ij} obey the rule of gauge transformation: $A_{ij} \rightarrow \lambda_i^{\alpha ij} A_{ij} \lambda_j^{\alpha ij}$, and then insert them in between matter fields

$$H = - \sum_{\langle ij \rangle} J_{\alpha ij} S_i^{\alpha ij} S_j^{\alpha ij} \rightarrow - \sum_{\langle ij \rangle} J_{\alpha ij} S_i^{\alpha ij} A_{ij} S_j^{\alpha ij}$$

- Define the electric fields E_i^μ

$$(E_i^\mu)^2 = 1 \quad , \quad E_i^{\alpha ij} A_{ij} E_i^{\alpha ij} = -A_{ij} \quad , \quad [E_i^\mu, E_j^\nu] = 0 \quad , \quad [E_i^\mu, S_j^\nu] = 0$$

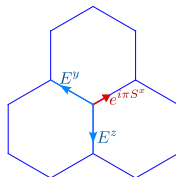
Anomalous 1-Form Symmetry

- Define the Gauss law (generators of gauge symmetries): Two independent generators

$$\begin{cases} G_k^{(x)} \equiv e^{i\pi S_k^x} E_k^y E_k^z \\ G_k^{(y)} \equiv e^{i\pi S_k^y} E_k^z E_k^x \\ G_k^{(z)} \equiv e^{i\pi S_k^z} E_k^x E_k^y = G_k^{(x)} G_k^{(y)} \end{cases}$$

Given a loop $\mathcal{L}$, suppose $k, \langle ij \rangle \in \mathcal{L}$

$$G_k^{(\hat{n}_k)} S_k^\mu G_k^{(\hat{n}_k)\dagger} = \lambda_k^\mu S_k^\mu, \quad \begin{aligned} G_i^{(\hat{n}_k)} A_{ij} G_i^{(\hat{n}_k)\dagger} &= \lambda_i^{\alpha_{ij}} A_{ij} \\ G_j^{(\hat{n}_k)} A_{ij} G_j^{(\hat{n}_k)\dagger} &= A_{ij} \lambda_j^{\alpha_{ij}} \end{aligned}$$



- Even-odd effect of the algebraic relation between Gauss law in different directions:

$$G^{(\mu)} G^{(\nu)} = (-1)^{2S} G^{(\nu)} G^{(\mu)}$$

Conclusion: When $S \in \mathbb{Z} + \frac{1}{2}$, the 1-form symmetry $\mathbb{Z}_2^{[1]}$ is **anomalous**.

Full Internal Symmetry Anomaly

The authors show that, for all S , there is **no pure 0-form** ($\mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$) **anomaly** or **mixed anomaly** between the 0-form symmetry and the 1-form symmetry $\mathbb{Z}_2^{[1]}$.

The classification of all anomalies associated with the $\mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)} \times \mathbb{Z}_2^{[1]}$ symmetry is (Sorry, I don't know why)

$$\mathbb{Z}_2^{17} = \mathbb{Z}_2^{10} \times \mathbb{Z}_2 \times \mathbb{Z}_2^6$$

- $\mathbb{Z}_2^{10}$: Containing only the **0-form** $\mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$ **symmetry**. 9 of them are classified using group cohomology (like the SPT classification)

$$H^4(\mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}, U(1)) = \mathbb{Z}_2^9$$

and the other one is beyond-cohomology anomaly (Vishwanath and Senthil, PRX 3, 011016 (2013))(?).

- $\mathbb{Z}_2$: Associated with the $\mathbb{Z}_2^{[1]}$ **1-form symmetry** (bosonic or fermionic statistics that the endpoints of strings obey)
- $\mathbb{Z}_2^6$: Mixed anomalies between the 0-form and 1-form symmetries, which stand for different **fractionalization patterns** of the $\mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$ symmetry on the endpoints of the strings, and are classified by

$$H^2(\mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}, \mathbb{Z}_2) = \mathbb{Z}_2^6$$

Full Internal Symmetry Anomaly

$$\mathbb{Z}_2^{17} = \underbrace{\mathbb{Z}_2^{10}}_{\text{pure 0-form anomaly}} \times \underbrace{\mathbb{Z}_2}_{\text{pure 1-form anomaly}} \times \underbrace{\mathbb{Z}_2^6}_{\substack{\text{mixed anomaly} \\ \text{(0-form symmetry fractionalization)}}}$$

- **No pure 0-form anomaly:** 0-form symmetries are on-site (Wen, PRB 99, 205139 (2019))(Wen, PRD 88, 045013 (2013))
- **No mixed anomaly:** $\mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)} \times \mathbb{Z}_2^T$ is not fractionalized non-trivially.

Sketch of the proof:

- Prove that the Majorana fermions in a spin-1/2 model do not fractionalize the 0-form symmetry.
- $2S$ copies of the spin-1/2 model $\Leftrightarrow$ Spin- S model
- A half-integer spin model has the same anomaly pattern as the spin-1/2 one.
- The integer spin models are non-anomalous since the classification is $\mathbb{Z}_2$.

Conclusion: The model only has a 1-form anomaly for general S .

Consequences of the Anomalies

$S \in \mathbb{Z} + \frac{1}{2}$: **Symmetry-enforced exotic quantum matter**

- Non-trivial anomaly: long-range entangled state
- 1-form anomaly: deconfined fermionic excitations

$S \in \mathbb{Z}$:

- Absence of anomaly: a $\mathbb{Z}_2^{[1]} \times \mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$ symmetric short-ranged entangled ground state is possible.
- Bosonic endpoints could condense, giving rise to no deconfined fractional excitation

Robustness:

- The discussions are **purely based on symmetries and anomalies**: The conclusion generally holds if the perturbation respects the $\mathbb{Z}_2^{[1]} \times \mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$ symmetry.
- Local weak perturbations respecting $\mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$ are allowed since it cannot completely destroy $\mathbb{Z}_2^{[1]}$: Emergent 1-form symmetry.

Conclusion

- The internal symmetries and their anomalies in the spin- S Kitaev model are discussed in a general fashion.
- Even-odd effect: The 1-form symmetry is anomalous if $S \in \mathbb{Z} + \frac{1}{2}$.
- There is only a pure 1-form anomaly.
- The result is robust if the $\mathbb{Z}_2^{[1]} \times \mathbb{Z}_2^T \times \mathbb{Z}_2^{(x)} \times \mathbb{Z}_2^{(y)}$ is respected.